

Video Transcript

Video 10: Connecting and Writing to a Database Using MongoDB (10:37)

So, now let's go ahead and take a look at the file for Mongo. And as you can see there that is pretty similar to the one that we just looked at for MySQL. However, it is simpler because Mongo requires less setup and configuration. We make the connection to the database, we get a handle on the database app, and configuration of the database in the document world. Same goes for the table. And then we write into that table. Notice that in this case we're passing in that new information. It takes a parameter here, and that parameter then is iterated through, and then we push as you can see here on the update.

We also have an additional function below it to be able to read the data to confirm that those updates, those writes, are in fact taking place. And then just like in MySQL, I have a last function there to be able to clear out the data in the database, on the MongoDB database. So, that was MySQL and MongoDB specific files. Now we're going to take a look at the scheduler. This is the file that holds the time loop. And as you can see there, this is the file, the MySQL DB and MongoDB files. We will use these here to be able to access those functions that we just overviewed. There's some additional things that we have here right below. Note that we have, and these are in addition to the time loop that starts here on line 23. We have a function to be able to clear out all of the data that we have in both of those databases.

And as you can see there that simply calls the functions that are part of MySQL and of Mongo. Below it, we have the deployment that we pass in at the command line and if we match that argument, you can see there that it simply clears out the data that we have in those databases. Now, here is the mean time loop, and we are going to overview here the functions. The very first one is one that simply writes to the console the data, and this is a good way to keep track of what we're reading and also note that it sleeps for five seconds. We do this simply to be able to see the data that is being written to the console. Below it, we have MySQL write. This is the one that leverages that function within the MySQL file and it does it write. This is what starts everything when it comes to being able to track those changes.

The next one is Mongo, and you can see that Mongo starts off by reading from MySQL. Once it does, it writes the status using the status function to the terminal, and then it writes into the Mongo database. Below it, we have a function to verify. It reads from Mongo and then writes to the terminal once more. And below that at the very end of the file we have the time loop. Time loop is pretty simple. It starts off by doing a time stamp to the terminal. Then it calls MySQL, followed by Mongo, followed by the verification function. Then it starts a timer which simply calls itself the function that it's in. And so, we get a time loop. And on line 55 we simply start that loop running.

Now, the last file to overview is the containers file. And as you might recall, this is the one where we delete containers, create containers, and initialize the databases that require it. So, as you can see there on my note on line 10, this is really only for MySQL, MongoDB does not need it. So, the very first block is the block to delete containers. You can see there that we are running the commands and writing acknowledgments to the terminal. The second block is to create containers. The third block is the one that initializes the database, and that runs from line 27 all the way to line 57, actually no, all the way to line 62. The block below that is the one that is reading the argument, and the three blocks, conditional blocks that follow it are the ones that are matching the kind of argument that was passed in. So, the very first one deletes containers. The second one creates containers, and the very last one is the one that initializes the database. So, that was quite a bit.

With that in mind, let's go ahead and move to the terminal and run this code. Let's start off here by listing our files, and then we're going to go ahead and run the containers file to be able to create our databases. And as you can see there, I have just opened the dashboard for Docker, so we can see when those containers are created. I'm going to go ahead and run the containers file and I'm going to use the argument for create. So, I'm going to go ahead and enter 'return'. And as you can see there, it starts off by pulling the images. I have cleared out my machine to make sure that it was clean. So, none of those images for those databases exist. And you can see there that one of the first one or the first one that was created there was MySQL, and now that is being followed by the Mongo image being pulled.

You can see there the process and the feedback that we get on the console. So, at this point, as you can see there, we have created both of those databases, and if we wanted to, we could go

ahead and delete them. So, let's go ahead and delete, and this will take a bit. There's quite a bit there to remove. You can see how the dashboard gets updated, and then we can go ahead and create it. So, as you can see there, this is pretty convenient. This is very fast. It's much easier than doing it by hand. And so, this is something that you can see there the advantage of.

So, let's go ahead and do the last thing, which is initialize that database. We want to be able to initialize MySQL DB. And I'm going to go ahead and run that command. And now that has been initialized. There has been a new database that has been added, the Pluto database. And inside of it, a table has been created, the table for posts. So, at this point, let's go ahead and clear the screen. And we're now going to run our scheduler. So, this is our main loop. This will start off populating MySQL. Those changes will be detected and then pushed into MongoDB, and we should see confirmations on the console. So, let's go ahead and run that. So, this would be Python 3 scheduler. And as you can see there, immediately we get the time stamp to the console.

You can see there that that data now has been copied into or has been updated. We have detected the change in data and, we have propagated it onto MongoDB. And you can see there the time stamps that we're getting now in the second loop or in the second time through that loop. You can see there that we have now two rows in MySQL, and you also have two rows within Mongo. And you can see there that the time stamps are being put into the terminal as we expect. Now, this is moving a little bit slow, and I'll show you how to make it faster. So, I'm going to go ahead and interrupt it here. As you can see there it complains, but that's ok. As you can see here, I am in the scheduler file and I'm going to go ahead and make the time loop go faster every five seconds, and the status pause, I'm going to make two seconds.

So, now I'm going to go ahead and restart. And as you can see there, I have just removed the dashboard because we don't need it anymore of Docker containers. I'm going to go ahead and clear here my terminal and then I'm going to go ahead and run the scheduler once more. As you can see there, we are reading the data in MySQL, then the data in Mongo. Then we should get the time stamp. As you can see there, we got it. And you can see there that that is now moving faster. Note that as time goes by, the very first row that is read is different. And you can see there that the later timestamps are being pushed out of those five that are being selected. So, go ahead and take a look at the functioning, running program here. And as you start to get some

results, make sure that you can match the logic that is being captured, and the way in which information is being propagated in this program.

Now, there's one more thing here to demonstrate. I'm going to go ahead and stop the program. And there was an option, if you remember, to be able to clear the data within both of those databases. I'm going to go ahead and run it. And as you can see there, we have cleared all of the information. If I was to run this scheduler again, you can see there that we start with empty databases. Again, a convenient thing to do when you're testing logic and you want to start over frequently. You'll want to write yourself that function to be able to clear out the data and start clean once again. Now, you can also delete the containers, but that takes a little bit longer, as you can see. Delete the container, delete. We're moving entire databases, and so it is more convenient often when it comes to databases to simply clear out the tables, clear out the databases, because that is a much quicker operation. So, now it is your turn.

